

Fermat's Little Theorem

Beulah Evanjalin

Contents

1	Introduction	3
1.1	Historical Context	3
1.2	Statement of the Theorem	3
2	Classical Proofs	4
2.1	Euler's Proof (1736)	4
2.2	Induction Proof	5
3	Combinatorial Proofs	6
3.1	Necklace Proof (Golomb, 1956)	6
3.2	String Rotation Proof	7
4	Group Theory Proofs	8
4.1	Lagrange's Theorem Approach	8
4.2	Cosets and Group Actions	9
5	Modern and Advanced Proofs	10
5.1	Field Theory Proof	10
5.2	Geometric Proof	11
6	Applications and Extensions	12
6.1	Computational Applications	12
6.1.1	Modular Exponentiation	12
6.1.2	Primality Testing	12
6.2	Cryptographic Applications	12
6.2.1	RSA Encryption	12
6.3	Theoretical Extensions	13
6.3.1	Euler's Theorem	13
6.3.2	Wilson's Theorem	13

1 Introduction

Pierre de Fermat's "Little Theorem" stands as one of the most elegant and powerful results in elementary number theory. First stated by Fermat in 1640 in a letter to Frénicle de Bessy, this theorem reveals a profound connection between prime numbers and modular arithmetic that continues to find applications in modern cryptography, primality testing, and theoretical mathematics.

1.1 Historical Context

The theorem emerged during the golden age of number theory, when mathematicians like Fermat, Euler, and Gauss were laying the foundations of modern algebra. While Fermat claimed to possess a proof that was "too long" to include in his correspondence, a recurring theme in his mathematical communications, the first published proof came nearly a century later from Leonhard Euler in 1736.

1.2 Statement of the Theorem

Theorem 1.1 (Fermat's Little Theorem - General Form). *If p is a prime number and a is any integer, then*

$$a^p \equiv a \pmod{p}$$

Theorem 1.2 (Fermat's Little Theorem - Coprime Form). *If p is a prime number and a is an integer such that $\gcd(a, p) = 1$, then*

$$a^{p-1} \equiv 1 \pmod{p}$$

These two formulations are equivalent: the first immediately implies the second by dividing both sides by a (valid since $\gcd(a, p) = 1$), while the second implies the first by multiplying both sides by a .

2 Classical Proofs

2.1 Euler's Proof (1736)

The most widely taught proof, attributed to Euler, uses the properties of multiplication modulo a prime.

Euler's Proof. Let p be prime and a an integer with $\gcd(a, p) = 1$. Consider the sequence:

$$S = \{a, 2a, 3a, \dots, (p-1)a\}$$

We claim that when reduced modulo p , this sequence is a permutation of $\{1, 2, 3, \dots, p-1\}$.

Step 1: No element of S is congruent to 0 modulo p

Since $\gcd(a, p) = 1$ and $1 \leq k \leq p-1$, we have $\gcd(ka, p) = 1$, so $ka \not\equiv 0 \pmod{p}$.

Step 2: All elements of S are distinct modulo p

Suppose $ia \equiv ja \pmod{p}$ for some $1 \leq i < j \leq p-1$. Then:

$$p \mid (j-i)a$$

Since $\gcd(a, p) = 1$, by Euclid's lemma, $p \mid (j-i)$. But $0 < j-i < p$, which is impossible for a prime p . Therefore, all elements are distinct.

Step 3: Completing the proof

Since S modulo p is a rearrangement of $\{1, 2, \dots, p-1\}$:

$$a \cdot 2a \cdot 3a \cdots (p-1)a \equiv 1 \cdot 2 \cdot 3 \cdots (p-1) \pmod{p}$$

This gives us:

$$a^{p-1} \cdot (p-1)! \equiv (p-1)! \pmod{p}$$

Since p is prime, $\gcd((p-1)!, p) = 1$, so we can cancel $(p-1)!$ from both sides:

$$a^{p-1} \equiv 1 \pmod{p}$$

□

Example 2.1 (Verification with $p = 7$, $a = 3$). Consider $S = \{3, 6, 9, 12, 15, 18\}$. Reducing modulo 7:

$$3 \equiv 3, \quad 6 \equiv 6, \quad 9 \equiv 2, \quad 12 \equiv 5, \quad 15 \equiv 1, \quad 18 \equiv 4$$

So $\{3, 6, 2, 5, 1, 4\}$ is indeed a rearrangement of $\{1, 2, 3, 4, 5, 6\}$. The product gives us:

$$3^6 \cdot 6! \equiv 6! \pmod{7}$$

Therefore $3^6 \equiv 1 \pmod{7}$, and indeed $3^6 = 729 \equiv 1 \pmod{7}$.

2.2 Induction Proof

This elegant proof demonstrates the power of mathematical induction in number theory.

Proof by Induction. We prove the general form: $a^p \equiv a \pmod{p}$ for all integers a .

Base Case: For $a = 0$: $0^p = 0 \equiv 0 \pmod{p}$

For $a = 1$: $1^p = 1 \equiv 1 \pmod{p}$

Inductive Step: Assume $k^p \equiv k \pmod{p}$. We must show $(k+1)^p \equiv k+1 \pmod{p}$.

Using the binomial theorem:

$$(k+1)^p = \sum_{i=0}^p \binom{p}{i} k^i = 1 + pk + \binom{p}{2} k^2 + \cdots + \binom{p}{p-1} k^{p-1} + k^p$$

Key Lemma: For prime p and $1 \leq i \leq p-1$, p divides $\binom{p}{i}$.

Proof of Lemma.

$$\binom{p}{i} = \frac{p!}{i!(p-i)!} = \frac{p \cdot (p-1)!}{i!(p-i)!}$$

Since p is prime and $1 \leq i \leq p-1$, neither $i!$ nor $(p-i)!$ contains p as a factor. Therefore, p divides the numerator but not the denominator, so $p \mid \binom{p}{i}$. $\square$

Returning to our main proof:

$$(k+1)^p \equiv 1 + 0 + 0 + \cdots + 0 + k^p \equiv 1 + k \pmod{p}$$

By the inductive hypothesis, $k^p \equiv k \pmod{p}$, so:

$$(k+1)^p \equiv 1 + k = k + 1 \pmod{p}$$

By induction, the theorem holds for all non-negative integers. Extension to negative integers follows from the fact that if $a^p \equiv a \pmod{p}$, then $(-a)^p \equiv -a \pmod{p}$ when p is odd. $\square$

3 Combinatorial Proofs

3.1 Necklace Proof (Golomb, 1956)

This beautiful combinatorial proof uses the concept of circular arrangements to demonstrate Fermat's Little Theorem.

Necklace Proof. Consider necklaces made with p beads, where each bead can be coloured in one of a different colours.

Step 1: Total colourings There are a^p ways to colour the necklaces when considering them as linear arrangements.

Step 2: Monochromatic necklaces Exactly a of these necklaces use only one colour (one for each colour).

Step 3: Rotational equivalence For the remaining $a^p - a$ necklaces that use at least two colours, we consider rotational equivalence. Two necklaces are equivalent if one can be obtained from the other by rotation.

Key Observation: If a necklace uses at least two colours, then its p rotations are all distinct. This is because if some rotation gives the same necklace, then the pattern would repeat with period $d < p$, where d divides p . Since p is prime, this forces $d = 1$, meaning all beads have the same colour, contradicting our assumption.

Step 4: Counting equivalence classes The $a^p - a$ non-monochromatic necklaces form equivalence classes of size p each. Therefore:

$$p \mid (a^p - a)$$

This proves $a^p \equiv a \pmod{p}$. □

Example 3.1 (Application with $a = 2$, $p = 5$). We have $2^5 = 32$ total colourings and 2 monochromatic necklaces. The remaining 30 necklaces form $30/5 = 6$ equivalence classes under rotation.

Indeed, $32 - 2 = 30 \equiv 0 \pmod{5}$, confirming $2^5 \equiv 2 \pmod{5}$.

3.2 String Rotation Proof

Another combinatorial approach uses the properties of string rotations.

String Rotation Proof. Consider all strings of length p over an alphabet of a symbols.

Total strings: a^p

Constant strings: a strings where all symbols are identical

Non-constant strings: $a^p - a$ strings using at least two different symbols

For any non-constant string S , define its **period** as the smallest positive integer d such that S is invariant under a cyclic shift by d positions.

Claim: If S is non-constant, its period is exactly p .

Proof of Claim. Suppose the period is $d < p$. Since rotations preserve the period, d must divide p . But p is prime, so $d = 1$. This means S is invariant under any single-position shift, implying all symbols are identical, contradicting that S is non-constant. $\square$

Therefore, each non-constant string generates exactly p distinct rotations, partitioning the $a^p - a$ non-constant strings into groups of size p .

Hence $p \mid (a^p - a)$, proving Fermat's Little Theorem. $\square$

4 Group Theory Proofs

4.1 Lagrange's Theorem Approach

This proof demonstrates the power of abstract algebra in number theory.

Group Theory Proof. Consider the multiplicative group of units modulo p :

$$U_p = \{1, 2, 3, \dots, p-1\}$$

under multiplication modulo p .

Step 1: U_p is indeed a group

- **Closure:** If $\gcd(a, p) = \gcd(b, p) = 1$, then $\gcd(ab, p) = 1$
- **Associativity:** Inherited from integer multiplication
- **Identity:** $1 \in U_p$
- **Inverses:** For each $a \in U_p$, there exists b such that $ab \equiv 1 \pmod{p}$ (by Bézout's identity)

Step 2: Apply Lagrange's theorem

For any $a \in U_p$, the cyclic subgroup $\langle a \rangle = \{1, a, a^2, \dots, a^{k-1}\}$ has order k , where k is the smallest positive integer such that $a^k \equiv 1 \pmod{p}$.

By Lagrange's theorem, k divides $|U_p| = p-1$.

Therefore, $a^{p-1} = (a^k)^{(p-1)/k} \equiv 1^{(p-1)/k} \equiv 1 \pmod{p}$. □

4.2 Cosets and Group Actions

A more sophisticated group-theoretic approach uses the concept of cosets.

Coset Proof. Let $G = (\mathbb{Z}/p\mathbb{Z})^*$ be the multiplicative group of units modulo p , and let $a \in G$.

Consider the map $\phi_a : G \rightarrow G$ defined by $\phi_a(x) = ax \bmod p$.

Step 1: ϕ_a is a bijection Since a has a multiplicative inverse in G , the map ϕ_a is invertible, hence bijective.

Step 2: Product preservation

$$\prod_{x \in G} x = \prod_{x \in G} \phi_a(x) = \prod_{x \in G} (ax) = a^{|G|} \prod_{x \in G} x$$

Step 3: Cancellation Since $\prod_{x \in G} x = (p-1)!$ and $\gcd((p-1)!, p) = 1$, we can cancel this product:

$$1 \equiv a^{|G|} = a^{p-1} \pmod{p}$$

□

5 Modern and Advanced Proofs

5.1 Field Theory Proof

This proof uses the structure of finite fields.

Field Theory Proof. The ring $\mathbb{Z}/p\mathbb{Z}$ is a field when p is prime. Consider the polynomial:

$$f(x) = x^p - x \in \mathbb{F}_p[x]$$

Step 1: Derivative analysis

$$f'(x) = px^{p-1} - 1 \equiv -1 \pmod{p}$$

Since $f'(x) \neq 0$, the polynomial $f(x)$ has no repeated roots.

Step 2: Counting roots The polynomial $f(x) = x^p - x = x(x^{p-1} - 1)$ has degree p , so it has at most p roots in $\mathbb{F}_p$.

Step 3: Finding all roots Every element $a \in \mathbb{F}_p$ satisfies $a^p = a$ by the field structure (this requires deeper field theory), so all p elements of $\mathbb{F}_p$ are roots.

Therefore, $a^p \equiv a \pmod{p}$ for all integers a . □

5.2 Geometric Proof

This innovative proof uses geometric intuition.

Geometric Proof. Consider the p -dimensional unit hypercube $[0, 1]^p$ and partition it into a^p unit subcubes of side length $1/a$.

Each subcube can be labeled with a p -tuple $(c_1, c_2, \dots, c_p)$ where $c_i \in \{0, 1, \dots, a-1\}$.

The transformation: Define $T : (c_1, c_2, \dots, c_p) \mapsto (c_2, c_3, \dots, c_p, c_1)$ (cyclic rotation).

Fixed points: A tuple is fixed by T if and only if $c_1 = c_2 = \dots = c_p$. There are exactly a such tuples.

Orbit analysis: For non-fixed tuples, T has order p (since p is prime), so each orbit has size p .

Therefore, the $a^p - a$ non-fixed tuples are partitioned into orbits of size p , proving $p \mid (a^p - a)$. $\square$

6 Applications and Extensions

6.1 Computational Applications

6.1.1 Modular Exponentiation

Fermat's Little Theorem dramatically simplifies modular exponentiation computations.

Example 6.1 (Computing $7^{100} \bmod 13$). Since 13 is prime and $\gcd(7, 13) = 1$:

$$7^{12} \equiv 1 \pmod{13}$$

Now, $100 = 12 \cdot 8 + 4$, so:

$$7^{100} = (7^{12})^8 \cdot 7^4 \equiv 1^8 \cdot 7^4 \equiv 7^4 \pmod{13}$$

Computing: $7^2 = 49 \equiv 10 \pmod{13}$ and $7^4 = 10^2 = 100 \equiv 9 \pmod{13}$.

Therefore, $7^{100} \equiv 9 \pmod{13}$.

6.1.2 Primality Testing

The Fermat primality test uses the contrapositive of Fermat's Little Theorem.

Theorem 6.1 (Fermat Compositeness Test). *If n is composite, then there exists some a with $1 < a < n$ such that $a^{n-1} \not\equiv 1 \pmod{n}$.*

However, there exist composite numbers called **Carmichael numbers** that satisfy $a^{n-1} \equiv 1 \pmod{n}$ for all a coprime to n . The smallest is $561 = 3 \cdot 11 \cdot 17$.

6.2 Cryptographic Applications

6.2.1 RSA Encryption

Fermat's Little Theorem (generalized as Euler's theorem) is essential for RSA cryptography.

In RSA with primes p and q , we use:

$$m^{ed} \equiv m \pmod{pq}$$

where $ed \equiv 1 \pmod{(p-1)(q-1)}$. This follows from Fermat's Little Theorem applied to each prime factor.

6.3 Theoretical Extensions

6.3.1 Euler's Theorem

Fermat's Little Theorem generalizes to:

Theorem 6.2 (Euler's Theorem). *If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$, where $\phi(n)$ is Euler's totient function.*

When $n = p$ is prime, $\phi(p) = p - 1$, recovering Fermat's Little Theorem.

6.3.2 Wilson's Theorem

An interesting corollary relates to factorials:

Theorem 6.3 (Wilson's Theorem). *$(p - 1)! \equiv -1 \pmod{p}$ if and only if p is prime.*

This follows from properties used in Euler's proof of Fermat's Little Theorem.